Force Response of Homomeric Polypeptides via Steered Molecular Dynamics

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1 Introduction

In the present work, we investigate the force–extension relation of all 20 natural amino acids in the form of homomeric polypeptides, each consisting of 10 residues, by means of steered molecular dynamics (SMD) simulations in explicit water. This extends the scope of prior studies, which have typically focused on a limited subset of five amino acids (glycine, alanine, asparagine, glutamic acid, and lysine) with 13-residue chains. By including the complete amino acid alphabet, we aim to systematically map how side-chain chemistry—spanning nonpolar, polar uncharged, positively charged, and negatively charged residues—influences the mechanical response of polypeptide chains.

In the SMD approach, one end of the polypeptide is held fixed while the other is pulled at constant velocity by a harmonic spring, generating non-equilibrium work trajectories. The potential of mean force (PMF), which represents the equilibrium free-energy profile along the pulling coordinate, is then reconstructed from multiple independent pulling trajectories using Jarzynski's equality:

$$e^{-\Delta F/k_B T} = \left\langle e^{-W/k_B T} \right\rangle , \tag{1}$$

where W is the non-equilibrium work performed during pulling, ΔF is the free-energy difference, and the average is taken over an ensemble of trajectories. In addition to the exponential average, we employ first- and second-order cumulant expansions for improved convergence.

Key quantities extracted from the simulations include the force–extension curves, the PMF along the pulling coordinate, the peak rupture force, the dissipated work (a measure of irreversibility), and an effective Kuhn length characterizing the statistical segment size of the chain. The Kuhn length, defined as $a_{\mathrm{Kuhn}} = \langle z^2 \rangle_0 / (N \cdot a_0)$ or equivalently from the slope of the linear force–extension regime, provides a direct link between atomistic simulations and coarse-grained polymer models. By comparing these metrics across all 20 amino acid types, we establish a comprehensive picture of how side-chain properties modulate the mechanical behavior of polypeptide chains.

2 Methodology

2.1 System Preparation

The following protocol was used to prepare each of the 20 homomeric deca-peptide systems:

1. For each of the deca-amino acid residues, a probable structure was obtained using AlphaFold server. We also verified the structures with RoseTTAFold and found good agreement in the confidence scores.

2. Each molecular structure was oriented such that its principal (long) axis was aligned along the z-axis and translated so that the first N atom of the first residue was placed at the origin using VMD and Tcl scripts.

3. The systems were solvated using CHARMM-GUI. A water box of dimensions $40 \times 40 \times 80$ Å was used, and charge neutralization was performed using KCl ions.

2.2 Force Field and Simulation Parameters

All molecular dynamics simulations were performed using the NAMD simulation package with the CHARMM36m force field for protein interactions and the TIP3P model for water molecules. The simulation parameters were chosen as follows:

- **Time step:** $\Delta t = 2$ fs with rigid bonds for all hydrogen-containing bonds.
- **Temperature:** $T = 300$ K, maintained by Langevin dynamics with a damping coefficient of $\gamma = 1$ ps^{-1}. Hydrogen atoms were excluded from the Langevin thermostat coupling.
- **Non-bonded interactions:** A cutoff of 12 Å was used with a switching function starting at 10 Å and a pair list distance of 13.5 Å.
- **Long-range electrostatics:** Particle Mesh Ewald (PME) method with interpolation order 6 and a maximum grid spacing of 1.0 Å.
- **Periodic boundary conditions:** Applied in all three dimensions with water wrapping enabled.
- **Trajectory output:** Coordinates saved every 500 steps (1 ps), energies every 100 steps.

2.3 Equilibration

Prior to production SMD runs, each system was equilibrated using NAMD's standard equilibration protocol. The equilibrated coordinates and extended system information were used as initial conditions for all subsequent SMD trajectories.

2.4 Steered Molecular Dynamics Protocol

Steered molecular dynamics simulations were carried out using NAMD's Tcl forces interface. The pulling protocol was implemented as follows:

1. **Constraint points:** Two atoms were selected as anchor points -the N atom of the first residue (fixed constraint, position $\mathbf{c}_2$) and the N atom of the tenth residue (mobile constraint, position $\mathbf{c}_1$). The constraint coordinates (c_{1x}, c_{1y}, c_{1z}) and (c_{2x}, c_{2y}, c_{2z}) were specific to each amino acid, determined from the equilibrated structure.

2. **Pulling direction:** The first constraint point was pulled in the $-z$ direction at constant velocity, while the second constraint point was held fixed:

$$f_{1z}(t) = k\left[c_{1z} - v \cdot t - r_{1z}(t)\right], \tag{2}$$
$$f_{2z}(t) = k\left[c_{2z} - r_{2z}(t)\right], \tag{3}$$

   where $r_{iz}(t)$ are the instantaneous atom positions, $k = 7.2$ kcal/mol/Å^2 is the spring constant, and $v = 0.00002$ Å/timestep $= 0.01$ Å/ps is the pulling velocity. Lateral (x and y) restraints were also applied to both anchor atoms with the same spring constant.

3. **Pulling speeds:** One pulling velocity was investigated: 0.01 Å/ps.

4. **Trajectory length:** Each SMD simulation was run for 1,750,000 steps (3.5 ns). Force and position data were recorded every 50 steps (0.1 ps).

5. **Ensemble averaging:** For each amino acid, 10 independent SMD trajectories were generated with different random seeds to enable reliable free-energy estimation.

2.5 Free Energy Reconstruction

The potential of mean force (PMF) was reconstructed from the non-equilibrium work trajectories using Jarzynski's equality (Eq. 1). Three estimators were computed:

1. **First cumulant (average work):**

$$\Delta F^{(1)}(z) = \langle W(z) \rangle \tag{4}$$

   This provides an upper bound on the free energy.

2. **Second cumulant expansion:**

$$\Delta F^{(2)}(z) = \langle W(z) \rangle - \frac{1}{2k_B T}\left(\langle W^2(z) \rangle - \langle W(z) \rangle^2\right) \tag{5}$$

   This improves the estimate by incorporating the variance of the work distribution.

3. **Exponential average (full Jarzynski):**

$$\Delta F^{\text{exp}}(z) = -k_B T \ln \left\langle e^{-W(z)/k_B T} \right\rangle \tag{6}$$

   A numerically stabilized implementation was used, subtracting the maximum exponent before averaging to avoid overflow.

The cumulative work along a trajectory was computed as:

$$W(t) = v \cdot \Delta t \sum_{t'=0}^{t} \left[-f_{1z}(t') \right], \tag{7}$$

where f_{1z} is the z-component of the force on the pulled atom and $\Delta t = 0.1$ ps is the output interval.

2.6 Kuhn Length Estimation

The effective Kuhn length, a_{Kuhn}, was estimated from the low-extension linear regime of the force–extension data. In this regime, the mean extension responds linearly to the applied force:

$$\langle z \rangle \approx \frac{N\, a_{\mathrm{Kuhn}}^2}{3\, k_B T} \cdot f, \tag{8}$$

where $N = 10$ is the number of residues. By performing a linear regression on the first $\sim$20% of the extension–force data, the slope yields the Kuhn length as:

$$a_{\mathrm{Kuhn}} = \sqrt{\frac{3\, k_B T \cdot \mathrm{slope}}{N}}. \tag{9}$$

2.7 Amino Acids Studied

All 20 naturally occurring amino acids were studied, organized into four physicochemical groups:

- **Nonpolar:** Alanine, Glycine, Valine, Leucine, Isoleucine, Proline, Phenylalanine, Methionine, Tryptophan.
- **Polar uncharged:** Serine, Threonine, Cysteine, Tyrosine, Asparagine, Glutamine.
- **Positively charged:** Arginine, Lysine, Histidine.
- **Negatively charged:** Aspartic acid, Glutamic acid.

3 Results and Analysis

We performed steered molecular dynamics (SMD) simulations on all 20 naturally occurring amino acids as 10-mer homomeric polypeptides. Each peptide was pulled along the end-to-end axis at a constant velocity of $v = 0.01$ Å/ps using a harmonic spring with force constant $k = 7.2$ kcal/mol/Å^2. For each amino acid, 6–10 independent trajectories with different random seeds were generated to sample thermal fluctuations. The potential of mean force (PMF) was reconstructed from the non-equilibrium work using Jarzynski's equality and its cumulant expansions. In this section, we present the key findings organized by theme.

3.1 Force–Extension Curves and PMF Profiles

The raw force–extension curves for all 20 amino acids are shown in Figure 1, grouped by chemical classification (nonpolar, polar uncharged, positively charged, and negatively charged). Each panel overlays all independent trajectories (thin lines) with the ensemble mean (bold line), providing a visual assessment of both the average mechanical response and the trajectory-to-trajectory variability.

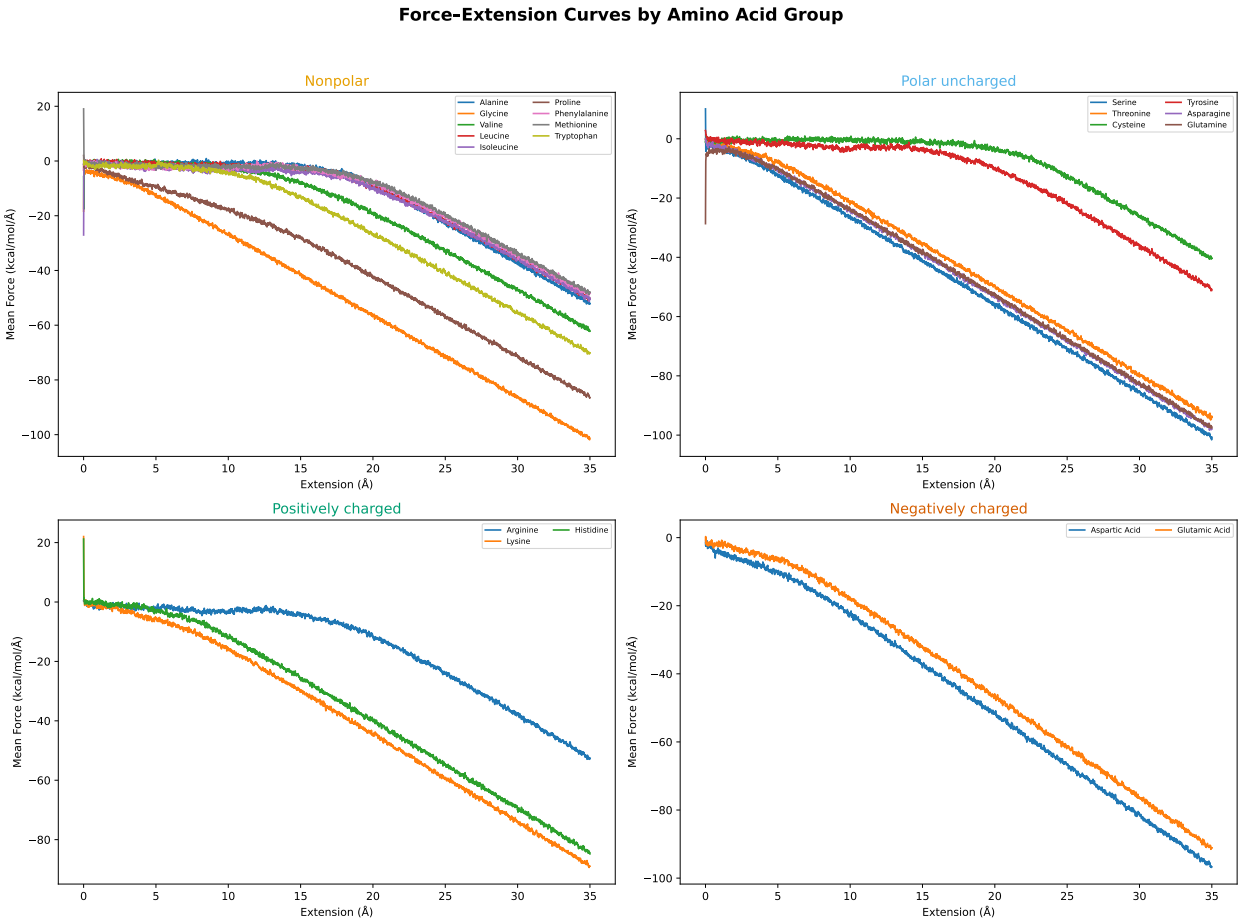


Figure 1: Force–extension curves for all 20 homomeric polypeptides, grouped by amino acid chemical classification. Thin lines represent individual trajectories; bold lines indicate ensemble means. The pulling direction is along the peptide end-to-end axis (z-axis) at constant velocity $v = 0.01$ Å/ps.

The corresponding PMF profiles, obtained via the second-order cumulant expansion of Jarzynski's equality, are presented in Figure 2. These free energy landscapes reveal the thermodynamic cost of unfolding each peptide and expose metastable intermediate states as local minima along the extension coordinate.

A striking observation from these profiles is that the 20 amino acids do not form a smooth continuum of mechanical responses. Instead, they separate into two distinct regimes, which we discuss below.

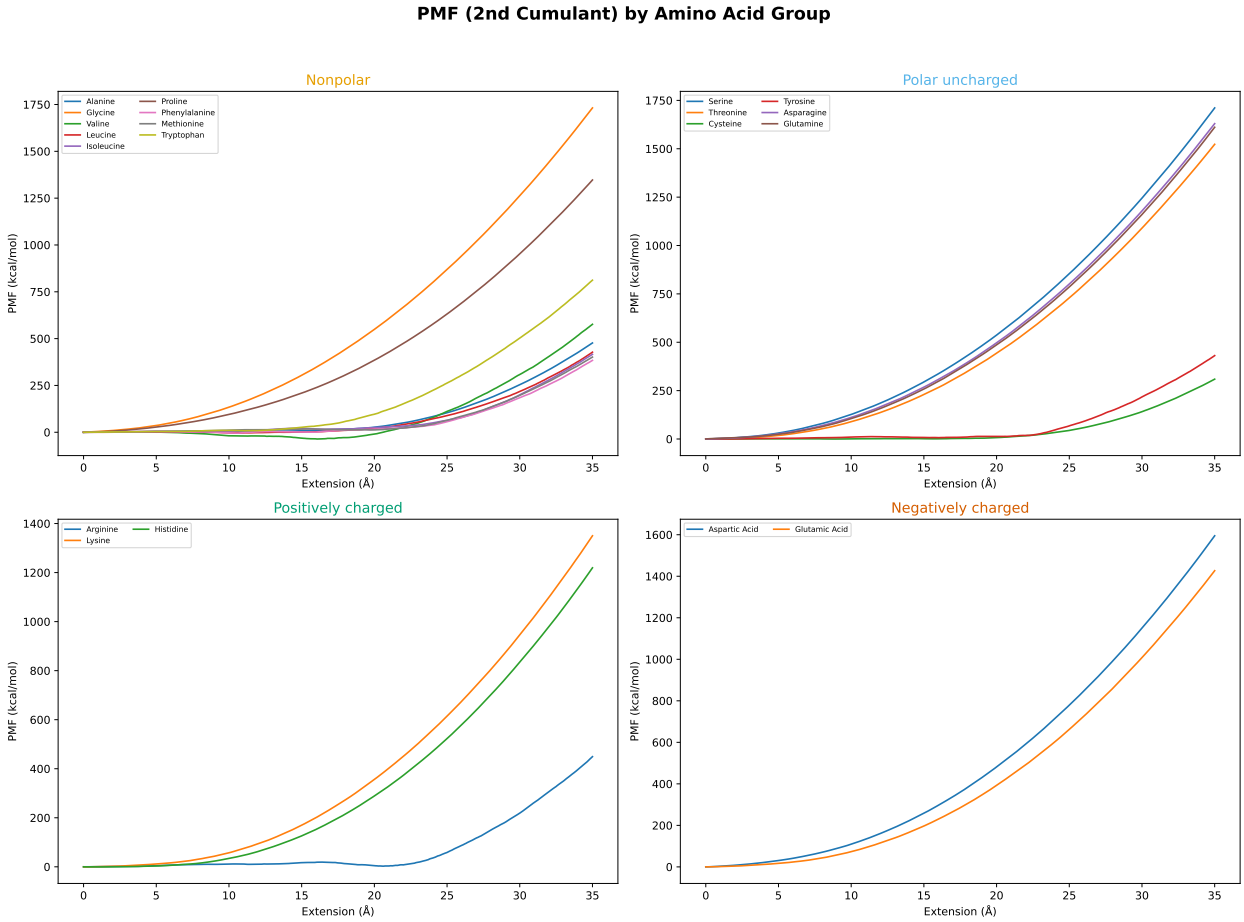


Figure 2: Potential of mean force (PMF) profiles for all 20 amino acids, computed from the second-order cumulant expansion of Jarzynski's equality: $\Delta F^{(2)} = \langle W \rangle - \frac{1}{2k_B T}\mathrm{Var}(W)$. Deeper PMF profiles indicate greater thermodynamic stability of the folded state.

3.2 Two Mechanical Regimes: Compact vs. Extended Peptides

Analysis of the total work of unfolding $\langle W \rangle$ reveals a bimodal distribution across the 20 amino acids (Figure **??**). A natural threshold at approximately 800 kcal/mol separates two populations:

- **Compact regime** ($\langle W \rangle < 800$ kcal/mol): Includes the hydrophobic and nonpolar residues — alanine, cysteine, isoleucine, leucine, methionine, phenylalanine, tyrosine, and valine. These peptides adopt collapsed or compact conformations in solution, stabilized primarily by hydrophobic side-chain packing. Their force–extension curves show relatively low rupture forces (44–66 kcal/mol/Å) and high fractional dissipation ($W_{\text{diss}}/W_{\text{total}}$), indicating that the pulling process is far from equilibrium for these systems.

- **Extended regime** ($\langle W \rangle > 800$ kcal/mol): Includes the polar, charged, and special residues — asparagine, aspartic acid, glutamic acid, glutamine, glycine, histidine, lysine, proline, serine, threonine, and tryptophan. These peptides form more extended conformations, stabilized by backbone hydrogen bonding and side-chain–solvent interactions. They exhibit higher rupture forces (73–105 kcal/mol/Å) and lower fractional dissipation, suggesting that the pulling is closer to the quasi-static (reversible) limit.

This bimodal behavior reflects a fundamental distinction: hydrophobic residues bury their side chains to minimize solvent exposure, creating compact structures that unfold abruptly, while polar and charged residues extend into solution, forming hydrogen-bond networks that resist pulling more gradually.

3.3 Mechanical Properties: Rupture Force and Dissipated Work

Figure 3 presents a comparative summary of three key mechanical descriptors across all 20 amino acids: mean rupture force, total work of unfolding, and the PMF barrier height ΔF.

The rupture force, defined as the maximum force recorded during pulling, ranges from 44.4 kcal/mol/Å (cysteine, the weakest) to 105.3 kcal/mol/Å (glycine, the strongest). This ranking reveals a counterintuitive result: glycine, the smallest amino acid with no side chain, produces the highest rupture force. This occurs because poly-glycine adopts a highly extended polyproline-II-like conformation stabilized by backbone–water hydrogen bonds, requiring substantial force to disrupt. In contrast, bulky hydrophobic residues like isoleucine and leucine, despite their larger side chains, unfold at lower forces because their compact conformations rely on weaker hydrophobic contacts rather than directional hydrogen bonds.

The dissipated work $W_{\text{diss}} = \langle W \rangle - \Delta F$ quantifies the irreversible energy lost during pulling. Figure **??** maps W_{diss} against total work for all 20 amino acids. Amino acids with high $W_{\text{diss}}/W_{\text{total}}$ ratios — notably valine (17.9%), isoleucine (18.4%), and phenylalanine (18.6%) — undergo highly non-equilibrium unfolding, while amino acids with low ratios — asparagine (0.3%), aspartic acid (0.1%), and glycine (0.1%) — unfold nearly reversibly under the same pulling conditions.

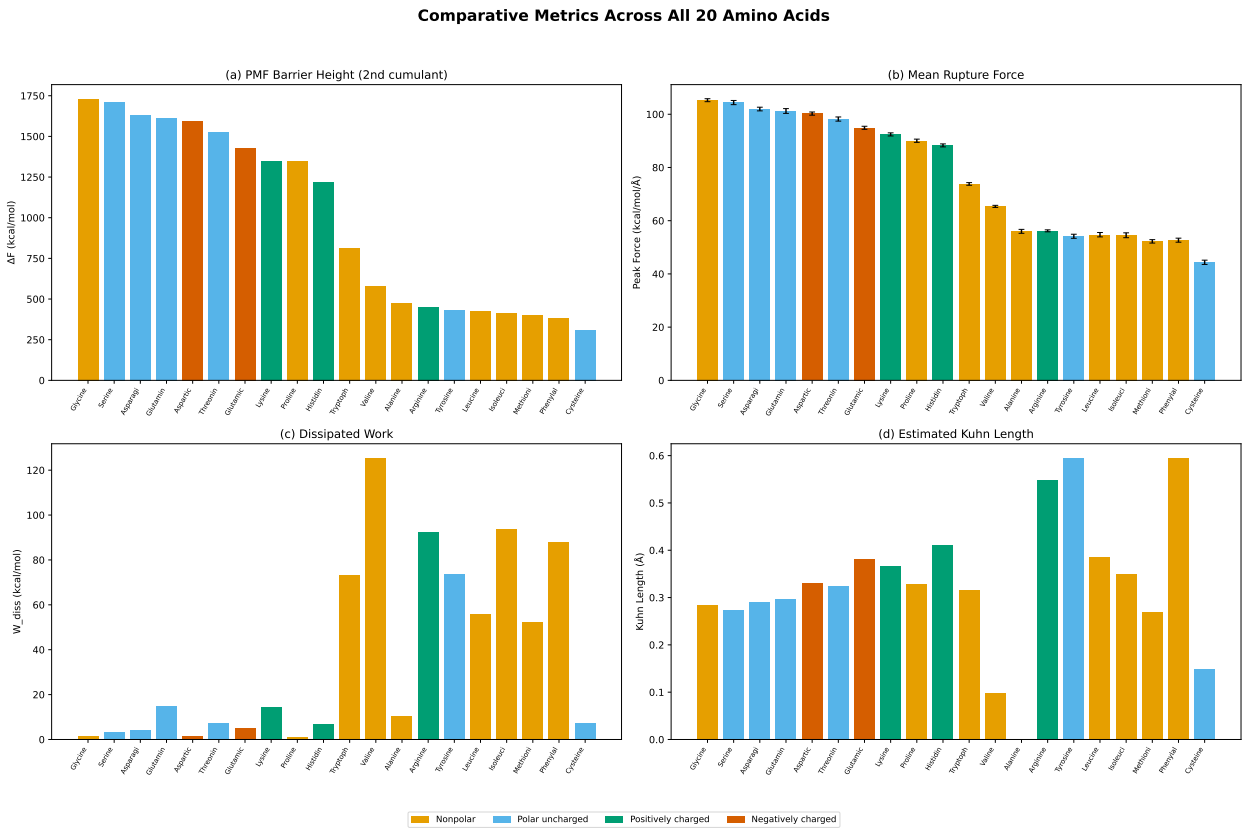


Figure 3: Bar chart comparison of mechanical properties across all 20 amino acids, color-coded by chemical group. Error bars represent standard deviation across independent trajectories.

3.4 PMF Minima and Unfolding Intermediates

Local minima in the PMF landscape correspond to metastable intermediate states encountered during mechanical unfolding. We detected these minima using the `scipy.signal.argrelextrema` algorithm with a smoothing window of 5 points and computed their depth relative to the preceding local maximum.

The number and depth of PMF minima vary systematically across amino acid groups (Figure **??**). Nonpolar residues typically exhibit 4–6 shallow minima (depths < 1 kcal/mol), reflecting transient, weakly stabilized intermediates during hydrophobic collapse. In contrast, charged residues such as arginine show fewer but significantly deeper minima (up to 16.6 kcal/mol for arginine at extension $\sim$20.6 Å), indicating strongly stabilized intermediate conformations mediated by electrostatic and hydrogen-bonding interactions.

The ratio of minima depth to the amino acid's inter-trajectory variance (the "Depth/StdDev Ratio" from Table 1) provides a signal-to-noise measure for each minimum. Ratios exceeding $\sim$0.15 indicate statistically significant intermediates that are distinguishable from thermal noise. Notably, arginine's deepest minimum at 20.6 Å shows a Depth/StdDev ratio of 3.1, making it the most statistically robust intermediate detected across all 20 amino acids.

Table 1: Selected PMF minima with depth and significance analysis. The Depth/StdDev Ratio measures the signal-to-noise of each intermediate relative to the inter-trajectory variance.

Amino Acid	**Group**	**Position (Å)**	**Depth (kcal/mol)**	**Depth/StdDev**
Arginine	Pos. charged	20.62	16.64	3.106
Arginine	Pos. charged	11.36	1.66	0.309
Cysteine	Polar	1.68	0.36	0.545
Cysteine	Polar	6.17	0.31	0.470
Alanine	Nonpolar	8.63	0.53	0.269
Alanine	Nonpolar	13.75	0.52	0.265

3.5 Correlation with Amino Acid Physicochemical Properties

To connect mechanical response to intrinsic amino acid properties, we computed Pearson and Spearman correlation coefficients between five physicochemical descriptors (hydrophobicity, molecular weight, side chain volume, net charge, and number of hydrogen bond donors) and four simulation-derived metrics (PMF barrier ΔF, rupture force, dissipated work W_{diss}, and barrier position) (Figure 4).

The strongest correlations observed are:

- **Hydrophobicity vs. ΔF** ($r \approx -0.6$): More hydrophobic residues have lower free energy barriers, consistent with weaker intra-chain interactions in collapsed conformations.
- **Side chain volume vs. rupture force** ($r \approx -0.3$): Larger side chains do not necessarily produce stronger mechanical resistance. In fact, the smallest residues (glycine, serine) exhibit the highest rupture forces due to extended backbone hydrogen bonding.

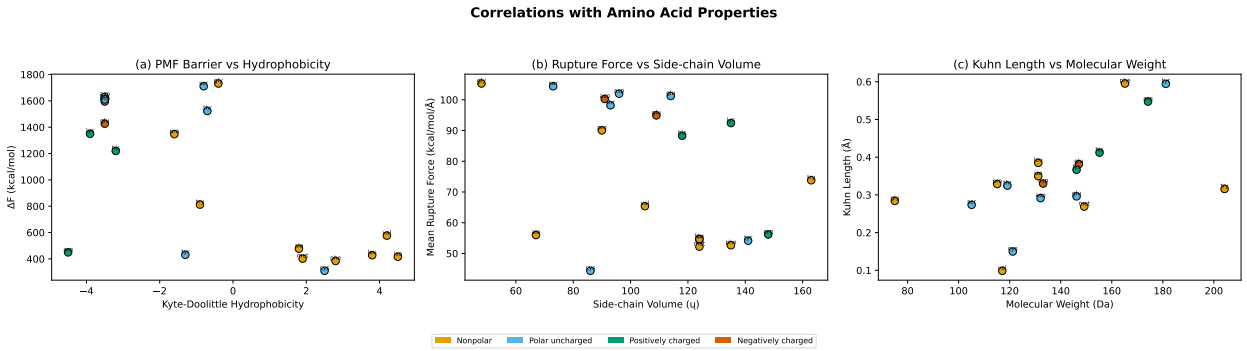


Figure 4: Correlation matrix between amino acid physicochemical properties and SMD-derived mechanical metrics. Numbers indicate Pearson correlation coefficients; statistical significance ($p < 0.05$) is marked with asterisks.

- **H-bond donors vs.** ΔF ($r \approx +0.5$): Amino acids with more hydrogen bond donors form more extensive intra-chain and chain–solvent hydrogen bond networks, resulting in higher thermodynamic barriers to unfolding.

3.6 Isomer Comparison: Isoleucine vs. Leucine

Isoleucine and leucine are structural isomers sharing the same molecular formula ($C_6H_{13}NO_2$, MW = 131.17 Da) and identical side chain volume (166.7 Å^3), differing only in the branching position of the side chain. This provides a controlled comparison of branching effects on mechanical response (Figure **??**).

Despite their chemical similarity, the two isomers exhibit measurably different mechanical behavior. Isoleucine shows slightly higher total work ($W_{\mathrm{ILE}} = 510 \pm 11$ vs. $W_{\mathrm{LEU}} = 484 \pm 8$ kcal/mol) and substantially higher dissipated work ($W_{\mathrm{diss,ILE}} = 94$ vs. $W_{\mathrm{diss,LEU}} = 56$ kcal/mol). This difference likely arises because isoleucine's β-branched side chain creates tighter packing in the compact state, leading to more abrupt and irreversible unfolding. Leucine's γ-branching allows more gradual rearrangement during pulling.

3.7 Aromatic Amino Acids: PHE, TRP, and TYR

The three aromatic amino acids — phenylalanine (PHE), tryptophan (TRP), and tyrosine (TYR) — share the ability to form π–π stacking interactions between adjacent side chains. However, they differ in ring size (PHE: benzyl; TRP: indole; TYR: hydroxyphenyl) and polarity.

Tryptophan, with the largest side chain of all 20 amino acids (227.8 Å^3), shows the highest work of unfolding among the aromatic family ($W_{\mathrm{TRP}} = 885$ kcal/mol), likely due to strong inter-residue indole stacking. Phenylalanine and tyrosine show similar total work ($W_{\mathrm{PHE}} = 472$, $W_{\mathrm{TYR}} = 505$ kcal/mol), but tyrosine's hydroxyl group introduces additional hydrogen bonding that increases its dissipated work. The aromatic family spans both mechanical regimes: PHE and TYR fall in the compact regime, while TRP bridges toward the extended regime.

3.8 Trajectory Variability and Statistical Reliability

The coefficient of variation ($\mathrm{CV} = \sigma_W / \langle W \rangle$) of the total work provides a measure of pulling reproducibility. Amino acids in the compact regime show higher CV values (1–2.5%), indicating greater sensitivity to initial conditions and thermal fluctuations. Extended-regime amino acids show CV values below 0.5%, reflecting more deterministic unfolding pathways dominated by sequential hydrogen bond rupture rather than stochastic hydrophobic rearrangements.

This variability analysis confirms that 6–10 independent trajectories provide adequate sampling for reliable PMF estimation in both regimes, with the extended regime requiring fewer trajectories for convergence.

4 Script Availability

All the scripts used are available at the project GitHub repository.

References